



Approval is a decision system— not an inbox.

Analysts re-key facts. Specialists reconstruct context. Sponsors assemble the final story.

01

FRAGMENTED

Evidence lives across files, messages and versions.

02

SERIAL

Every handoff waits for the previous inbox.

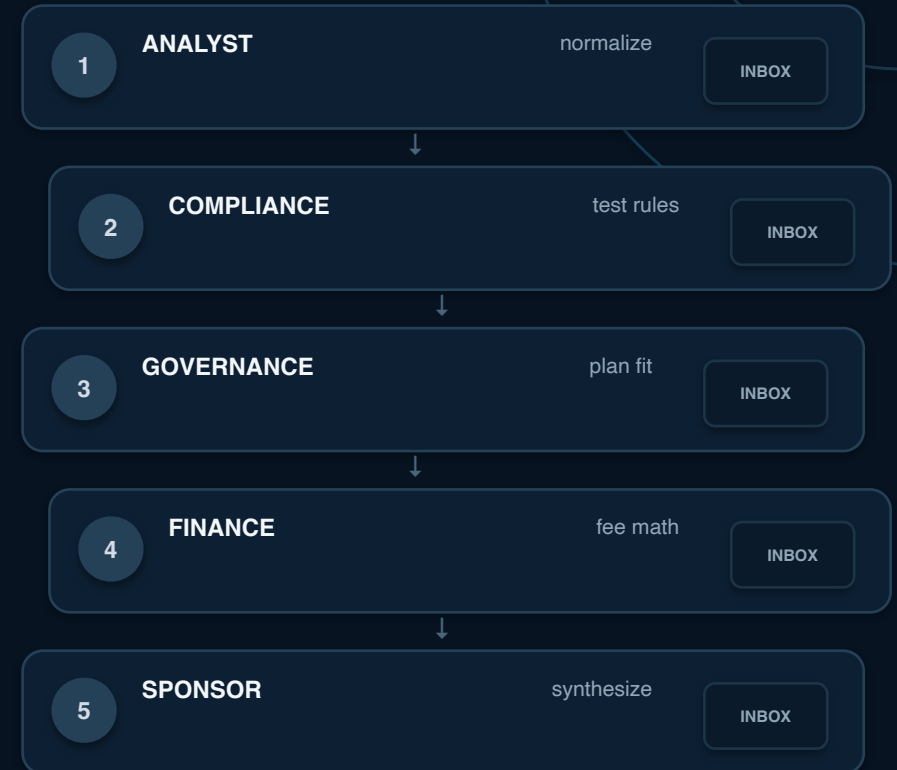
03

OPAQUE

The 'why' is rebuilt after the decision.

A fluent answer is not a control.

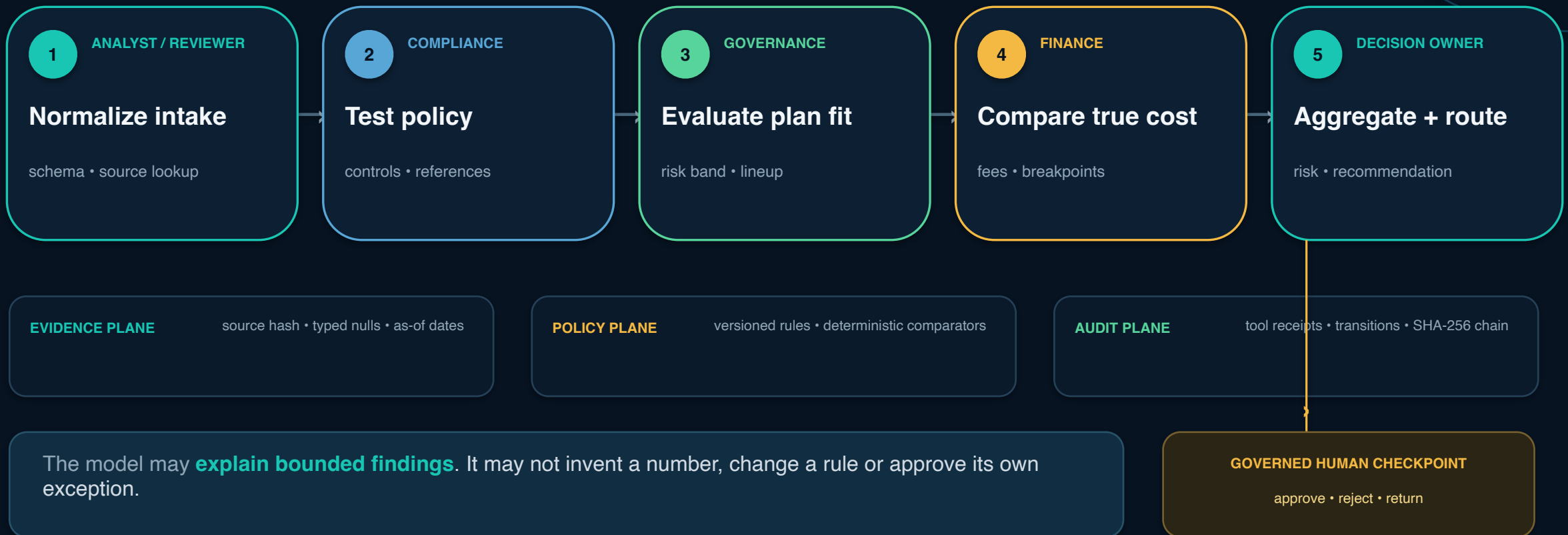
Fiducia creates one case, one evidence chain and one accountable next action.





Five bounded agents. One evidence chain.

LangGraph owns state and routing. Deterministic code owns validation, math and decision gates.





The demo proves behavior—not promises.

Run SUNX. Break DATA. Open the governed human checkpoint.



01 **GOLDEN PATH**
SUNX → APPROVE
no human touch

02 **RECOVERY**
DATA → retry 1/2
source recorded

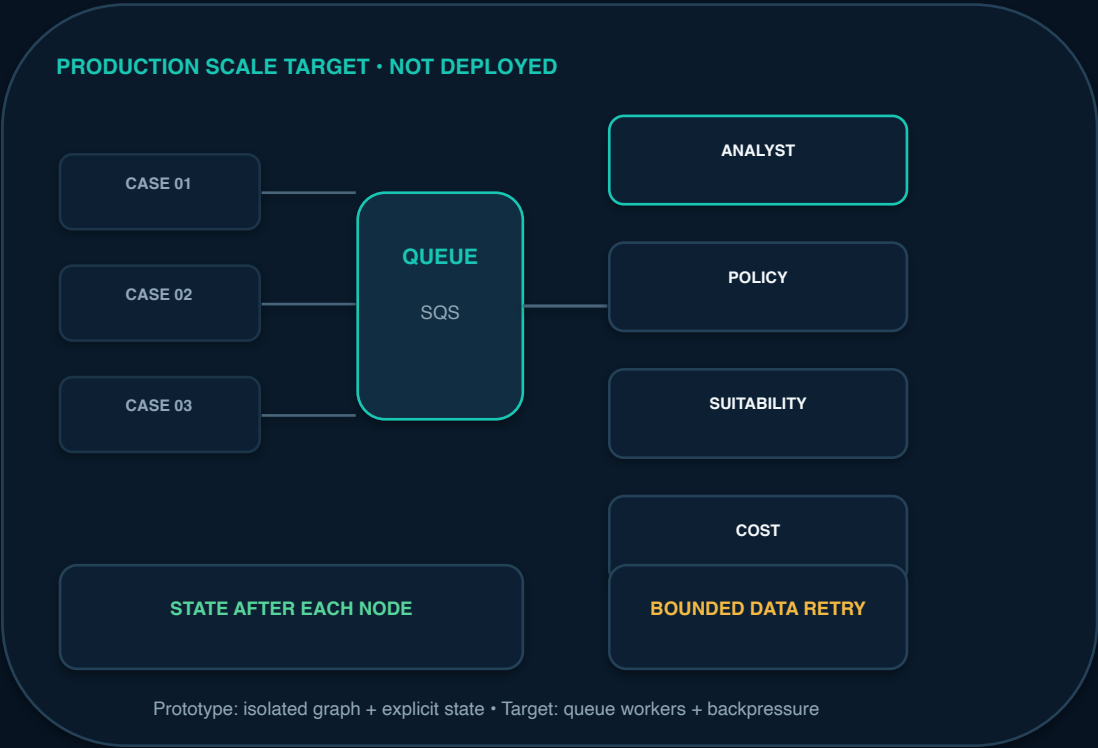
03 **CONTROL**
condition → pause
reason required

SHOW EVENTS, NOT HIDDEN CHAIN-OF-THOUGHT



Scale the workflow—never the uncertainty.

The prototype proves isolated case state and bounded retry; the production target adds queue-backed scale.



VALUE HYPOTHESES • MEASURE IN PILOT

FASTER DECISIONS

Target: parallel reviews

Measure: p50 / p95 cycle time

EXPERT CAPACITY

Exceptions—not re-keying

Measure: human touch-hours

CONTROL CONSISTENCY

One versioned policy

Measure: concordance / defects

AUDIT READINESS

Reconstructable evidence trace

Measure: packet reconstruction time

Design separates policy packs from prompts and model releases.



Earn autonomy one gate at a time.

Begin in shadow mode, measure safety and value, then widen only the scopes that prove both.

ILLUSTRATIVE • NOT A TIAA FORECAST

\$2.20M

gross annual value

\$1.25M

year-1 cost

75.8%

year-1 ROI

5.7 mo

simplified payback

5,000 cases × 8h × \$95 × 55% + rework value

Replace every assumption with shadow-pilot measurements.

PROPOSED MODULAR ADOPTION

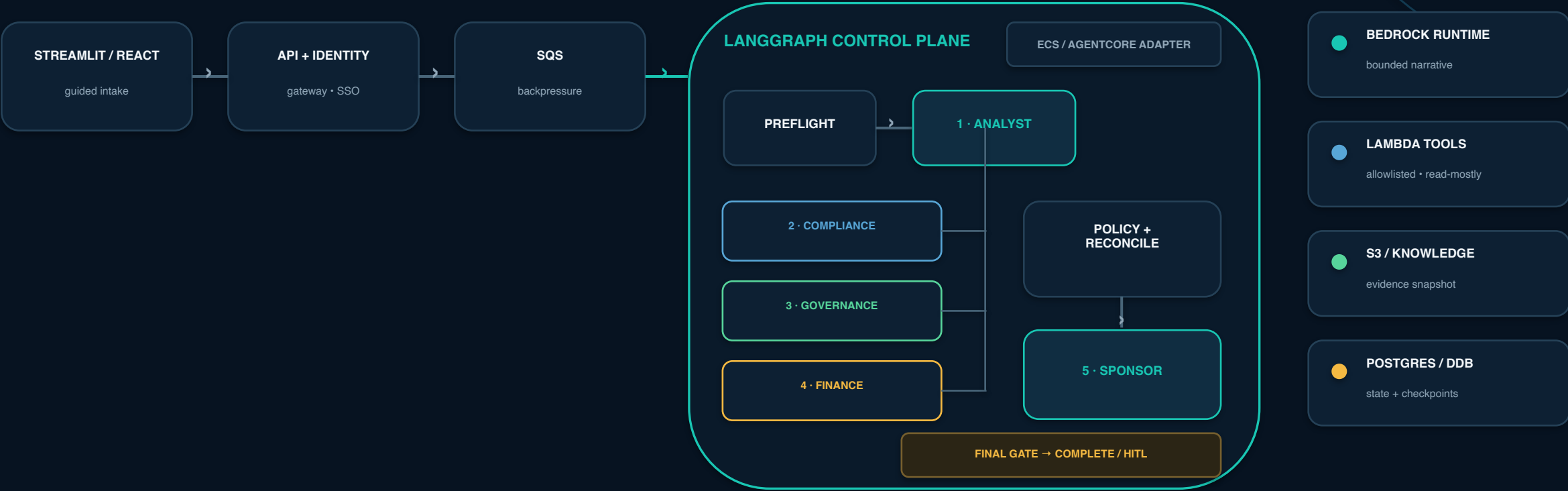


Target: faster decisions. Visible controls. Proof attached.



One orchestration authority. Many bounded capabilities.

Production target—not deployed locally: LangGraph lifecycle, Bedrock language and policy-as-code gates.



LOCAL → TARGET JSONL + SHA-256 chain → production S3 Object Lock

Target: CloudWatch • OpenTelemetry • cost / latency / safety metrics



Assume hallucinations. Remove their authority.

Every risky output must survive provenance, schema, deterministic checks and a bounded route.



BLANK

Invented missing fee

DETECT typed null + source hash

RESPOND retrieve twice → review

INJX

Prompt injection

DETECT input scanner + boundary

RESPOND isolate content → human

STALE

Wrong / stale rule

DETECT hash + effective date

RESPOND safe stop → approved pack

CONFX

Conflicting values

DETECT field reconciliation

RESPOND preserve both → escalate

UNITS

Math / unit error

DETECT Decimal + basis points

RESPOND ignore prose arithmetic

AUDIT

Audit commit fails

DETECT hash-chain verification

RESPOND no final recommendation

Five agents agreeing is not evidence. Independence comes from approved sources, deterministic tools and labeled adversarial cases.



Autonomy is earned by explicit gates.

Demo defaults are configurable illustrative policy—not statements of law.

AUTO-COMPLETE WORKFLOW

ALL must pass

- Recommendation = APPROVE
- Confidence ≥ 0.90
- Risk $\leq 24 / 100$
- > 5 bps from internal cap
- No hard stop / missing / conflict / injection
- All agent + audit events committed

RECOMMENDATION ONLY

HUMAN REVIEW

ANY one triggers

- Any non-APPROVE recommendation
- Confidence $< 0.85^*$
- Risk $\geq 40 / 100$
- Fee within ± 5 bps
- Missing after 2 / 2 retries
- Conflict, product scope or injection
- Hard stop or positive sales load

*0.85–0.89 cannot auto-complete; routing remains policy-controlled.

TOP JUDGE QUESTIONS

01 How do you prevent hallucinations?

We do not promise zero hallucinations. Typed nulls, source hashes, schema gates and deterministic math make model prose non-authoritative.

02 Is this truly autonomous?

The prototype autonomously prepares, tests, retries and routes a recommendation; it does not execute a financial decision.

03 How does it scale safely?

Today: isolated in-memory case state and bounded retry. Target: durable state and queue workers—after shadow safety and value gates pass.

DEMO: ENTERED ID + REASON + ATTESTATION · REJECT→APPROVE: DISTINCT 2ND ID